National Institutes of Health
Office of Dietary Supplements

Botanical Dietary Supplements Background Information

Fact Sheet for Consumers

Expand All

What is a botanical?

A botanical is a plant or plant part valued for its medicinal or therapeutic properties, flavor, and/or scent. Herbs are a subset of botanicals. Products made from botanicals that are used to maintain or improve health are sometimes called herbal products, botanical products, or phytomedicines.

In naming botanicals, botanists use a Latin name made up of the plant's genus and a term called the specific epithet. Together, this phrase represents the plant's species name. For example, the botanical black cohosh is known as *Actaea racemosa* L., where L stands for Linnaeus, who first described this plant. The Office of Dietary Supplements (ODS) fact sheets (https://ods.od.nih.gov/factsheets/list-all/) do not include such initials because they do not appear on the labels of most products used by consumers.

Can botanicals be classified as dietary supplements?

To be classified as a dietary supplement, a botanical must meet the definition of a dietary supplement as defined by Congress in the Dietary Supplement Health and Education Act of 1994. This act states that a dietary supplement is a product (other than tobacco) that

- Is intended to supplement the diet
- Contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, or other substances) or their constituents
- Is intended to be taken by mouth as a pill, capsule, tablet, or liquid
- Is labeled as a dietary supplement

How are botanicals commonly sold and prepared?

Botanicals are sold in many forms as both fresh and dried plant materials. For example, a supermarket's produce aisle carries fresh ginger root, while dried ginger root may be found in the dietary supplement aisle in capsule or tablet form, in tea bags, or as a liquid preparation. A group of chemicals or a single chemical may also be isolated from a botanical and sold as a dietary supplement, usually in tablet or capsule form. For example, phytoestrogens from soy products are sold as dietary supplements.

Common ways to prepare botanicals for use include teas, infusions, decoctions, tinctures, and extracts:

- A tea is made by adding boiling water to fresh or dried botanicals and steeping them. An infusion is a very strong tea. Teas and infusions may be consumed hot or cold.
- Some roots, bark, and berries require more forceful treatment to extract their desired constituents. They are added to water that is brought to a boil and then simmered at lower heat for several minutes. This also reduces the volume of liquid, producing a more concentrated preparation. Decoctions may be consumed hot or cold.
- A tincture is made when a botanical is soaked in a solution of alcohol and water. Tinctures are sold in liquid form and are used for concentrating and preserving a botanical. They are available in different strengths that are expressed as botanical-to-extract ratios (i.e., ratios of the weight of the dried botanical to the volume or weight of the finished product).
- An extract is made when the botanical is soaked in a specific liquid, such as water or alcohol, to extract the desired constituents. The extract can be used after soaking or it can be evaporated to make a dry extract for use in capsules or tablets.

Are botanical dietary supplements standardized?

Standardization is a process that manufacturers may use for extracts to ensure that all of their batches are similar. The standardization process involves identifying and measuring specific chemicals (also known as markers) and adjusting them to assure consistent amounts in each batch.

Ideally, the chemical markers chosen for standardization would also be the constituents that are responsible for a botanical's effect in the body. If such chemical markers were used, each batch of the product would have the same health effects. However, the constituents responsible for the effects of most botanicals are not known. For example, the sennosides in the botanical senna are responsible for its laxative effect, but many constituents may be responsible for valerian's relaxing effect.

U.S. law does not require dietary supplements to be standardized. In fact, there is no legal or regulatory definition of the term in the United States.

Are botanical dietary supplements safe?

Some people believe that products labeled natural are safe and good for them. This belief is not necessarily correct because the safety of a botanical depends on many things, such as its chemical makeup, how it works in the body, how it is prepared, and the amount used.

The actions of botanicals range from mild to powerful. A botanical with mild action might have subtle effects. Chamomile (https://www.nccih.nih.gov/health/chamomile) and peppermint, for example, are usually consumed in teas to help with digestion and are generally considered safe for most people. Some botanicals with mild action might need to be taken for weeks or months before their full effects are achieved. For example, valerian (https://ods.od.nih.gov/factsheets/Valerian-HealthProfessional/) might help users sleep better after a few weeks of use, but just one dose is rarely effective. In contrast, a

powerful botanical produces a fast result. Green tea (https://www.nccih.nih.gov/health/green-tea) (a natural source of caffeine) and yohimbe (https://www.nccih.nih.gov/health/yohimbe), for example, can have strong and immediate stimulant effects.

The dose and form of a botanical preparation also play important roles in its safety. Teas, tinctures, and extracts have different strengths from one another. For example, the same amount of a botanical may be contained in a cup of tea, a few teaspoons of tincture, or an even smaller amount of an extract. Also, different preparations have different amounts and concentrations of constituents extracted from whole botanicals. For example, peppermint tea is generally considered safe to drink, but peppermint oil is much more concentrated and can be toxic if used incorrectly.

Follow the manufacturer's suggested directions for using a botanical and do not exceed the recommended dose unless your health care provider directs otherwise. In fact, you should always talk with your health care provider about botanical and other dietary supplements that you are using or are thinking of using.

Does a label indicate the quality of a botanical dietary supplement product?

Determining the quality of a botanical dietary supplement product from its label is difficult. The degree of quality control depends on the manufacturer and others in the production process. The presence of terms such as standardized, for example, does not necessarily indicate that the product is of high quality.

The U.S. Food and Drug Administration (FDA) has established good manufacturing practices (GMPs) that dietary supplement manufacturers must follow to help ensure the identity, purity, strength, and composition of their dietary supplements. These GMPs can prevent the use of the wrong ingredient (or too much or too little of the right ingredient) and reduce the chance of contamination or improper packaging and labeling of a product. The FDA periodically inspects facilities that manufacture dietary supplements.

Several independent organizations offer quality testing and allow products that pass these tests to display a seal of quality assurance. These seals indicate that a product was properly manufactured, contains the ingredients listed on the label, and does not contain harmful levels of contaminants. However, these seals do not guarantee that a product is safe or effective. Organizations that offer quality testing include:

- ConsumerLab.com (https://www.consumerlab.com/)
- NSF International (https://www.nsf.org/knowledge-library/supplement-vitamin-certification)
- U.S. Pharmacopeia (https://www.quality-supplements.org/)

What methods are used to evaluate the health benefits and safety of a botanical dietary supplement?

Scientists can use many approaches to evaluate the potential health benefits and risks of botanical dietary supplements. For simple single-ingredient products, they can investigate the history of the botanical's use. They can also conduct laboratory studies of the botanical's effects on cell or tissue cultures or examine its effects in animals. Studies with people (for example, individual case reports, observational studies, and clinical trials) provide the most direct evidence of a botanical supplement's effects on health and how people are using it. These studies are important, especially for complex multi-ingredient products.

The amount of scientific evidence available on the health effects and safety of botanical ingredients varies widely. For example, scientists have conducted numerous studies (with mixed results) on the use of black cohosh to treat menopausal symptoms, including hot flashes and night sweats. On the other hand, very little research has been done on some botanical ingredients, such as astragalus (https://www.nccih.nih.gov/health/astragalus), to determine their value.

Through the Consortium for Advancing Research on Botanical and Other Natural Products (CARBON) Program, ODS promotes research on the safety, effectiveness, and mechanisms of action of botanical dietary supplements with high potential to benefit human health. The CARBON Program also supports the development of methods and resources to enhance the progress of this research.

Medicines must be evaluated for safety and efficacy and receive FDA approval before they can be sold or marketed, but dietary supplements, including botanicals, do not require FDA approval. The FDA requires supplement companies to have evidence that their products are safe, and claims on product labels must be truthful and not misleading.

What are some additional sources of information on botanical dietary supplements?

- For information about the safety and effectiveness of botanical and other dietary supplement ingredients, see the ODS Dietary Supplement Fact Sheets and the Herbs at a Glance (https://www.nccih.nih.gov/health/herbsataglance) series from the National Center for Complementary and Integrative Health.
- For general information about the use of dietary supplements, see Dietary Supplements: What You Need to Know.
- For regulatory information on dietary supplements, visit the FDA website (https://www.fda.gov/food/dietary-supplements).

Disclaimer

This fact sheet by the National Institutes of Health (NIH) Office of Dietary Supplements (ODS) provides information that should not take the place of medical advice. We encourage you to talk to your health care providers (doctor, registered dietitian, pharmacist, etc.) about your interest in, questions about, or use of dietary supplements and what may be best for your overall health. Any mention in this publication of a specific product or service, or recommendation from an organization or professional society, does not represent an endorsement by ODS of that product, service, or expert advice.

Glossary

amino acid
: A chemical building block of protein.

black cohosh
: A plant whose rhizome and root are used to relieve hot flashes and other symptoms of menopause. It is not known how black cohosh works or whether it acts like estrogen (a hormone needed to develop and maintain female sex characteristics and the growth of long bones). Historically, black cohosh has been used to treat many medical conditions. Also called black snakeroot, bugbane, bugwort, rattleroot, rattletop, rattleweed, and macrotys. Latin names: *Actaea racemosa* and *Cimicifuga racemosa*.

botanical
: Having to do with plants or plant parts, or dietary supplement products made from plants.

botanist
: A scientist who studies the biology of plants.

capsule
: A gelatin shell containing a dose of medicine, a vitamin, or other dietary supplement.

case report
: A detailed record of the diagnosis, treatment, and follow-up of an individual patient. Case reports also contain some information about the patient (such as age, gender, and ethnic origin).

cell
: The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

clinical trial
: A type of research study that uses volunteers to test the safety and efficacy (the ability to produce a beneficial effect) of new methods of screening (checking for disease when there are no symptoms), prevention, diagnosis, or treatment of a disease. Also called a clinical study.

constituent
: A component, part, or ingredient of a larger whole. For example, valerenic acid and valepotriate are constituents of the dietary supplement valerian.

consume
: To eat or drink.

decoction
: A substance made by simmering some types of roots, bark, and berries in water to extract their desired ingredients. It is simmered for a longer time than that needed to make tea and may be drunk hot or cold.

dietary supplement
: A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

digestion
: The process the body uses to break down food into simple substances for energy, growth, and cell repair.

dose
: The amount of medicine or other substance taken at one time or over a specific period of time.

effectiveness

In medicine, the ability of an intervention (for example, a drug, surgery, or a dietary supplement) to produce the desired beneficial effect under the usual conditions of care by a health care provider.

efficacy

In medicine, the ability of an intervention (for example, a drug, surgery, or dietary supplement) to produce the desired beneficial effect under the best possible conditions of care, such as a clinical trial.

evidence

Information used to support the use of a particular screening procedure, treatment, or preventive measure. In medicine, evidence needed to determine effectiveness is provided by laboratory research, clinical trials, and other studies.

extract

A substance made by soaking an herb in a liquid that removes specific types of chemicals. The liquid can be used as is or evaporated to make a concentrate or a dry extract for use in capsules or tablets.

Food and Drug Administration

FDA, Department of Health and Human Services. FDA is the Federal government agency responsible for ensuring that foods and dietary supplements are safe, wholesome and sanitary, and that drugs, medical devices, cosmetics, and food are honestly, accurately and informatively represented to the public. FDA regulates dietary supplements under a different set of regulations than those covering conventional foods and drug products (prescription and over-the-counter). The dietary supplement manufacturer is responsible for ensuring that a dietary supplement is safe before it is marketed. FDA is responsible for taking action against any unsafe dietary supplement product after it reaches the market. Generally, manufacturers do not need to get FDA approval before producing or selling dietary supplements.

genus

The name of a category that is part of the scientific classification of all organisms. Genus is located in the classification system after kingdom, phylum, class, order, and family and before the subclassification of species. Humans, for example, belong to the genus *Homo* and the species *Homo sapiens*.

ginger

The root of this plant has been used in cooking and in some cultures to treat nausea, vomiting, and certain other medical conditions. It is being studied in the treatment of nausea and vomiting caused by cancer chemotherapy. Latin name: *Zingiber officianale*.

herb

A plant used in cooking, in tea, and for medicinal purposes.

herbal

Having to do with or made from medicinal or edible plants.

hot flash

A sudden, temporary onset of body warmth, flushing, and sweating (often associated with menopause).

ingredient

In a dietary supplement, an ingredient is a component of the product, such as the main nutrient (vitamin, mineral, herb, amino acid, or enzyme) or any binder, color, filler flavor, or sweetener. In herbal supplements, the common name and Latin name (the genus and species) of the plant is given in the ingredient list. On a dietary supplement label, the ingredients are listed by weight, with the ingredient used in the largest amount first on the list and the ingredient used in the least amount at the end of the list.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a “supplement”; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

laboratory study

Research done in a laboratory. A laboratory study may use cells in test tubes or animals to find out if a drug, procedure, or other treatment is likely to be safe and useful. Laboratory studies usually take place before any testing is done in humans.

laxative

A substance that moves the bowels and relieves constipation.

medical history

Information about a person’s health, such as allergies, illnesses, surgeries, medications, immunizations, and the results of tests and physical exams. It may also include information about health habits, such as diet and exercise, and health information about current and past illnesses of one’s parents and other close family members.

medicinal

Having to do with the abilities of medicine to prevent and cure.

menopause

The time of life when a woman’s menstrual periods stop. A woman is in menopause when she hasn’t had a period for 12 months in a row. Also called “change of life.”

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

peppermint

A plant that has been used in traditional medicine in many parts of the world to relieve indigestion, cough, sore throat, headache, abdominal cramping, and gas. Also called brandy mint, lamb mint, and lammint. Latin name: *Mentha piperita*.

phytoestrogen

A weak estrogen-like substance found in some plants and plant products. Isoflavones, a type of phytoestrogen, are being studied in the prevention of osteoporosis, menopausal symptoms, and some types of cancer. Soybeans are a rich source of phytoestrogens.

phytomedicine

The use of herbs and other plants for their therapeutic or medicinal qualities. Also called phytotherapy.

preparation

A mixture made for medicinal use.

prevent

To stop from happening.

quality control

A system to ensure that consistency and uniformity are maintained in the manufacturing of a product.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

root

A part of a plant that is below ground.

senna

A plant used in traditional Chinese medicine and Ayurvedic medicine. The leaves are used to make a stimulant laxative that increases the frequency of bowel movements and relieves constipation. It is widely used in over-the-counter laxatives. Latin name: *Senna alexandrina*.

sennoside

The active ingredient in senna, a plant whose leaves are used to make a stimulant laxative that increases the frequency of bowel movements and relieves constipation.

solution

A liquid in which another substance has been dissolved or mixed.

soy

A plant that produces beans used in many food products. Soy products contain isoflavones (estrogen-like substances) that are being studied in the prevention of cancer, hot flashes that occur with menopause, and osteoporosis (loss of bone density). Also called soya and soybean. Latin name: *Glycine max*.

species

The name of a category that is part of the scientific classification of all organisms. The category species is located in the classification system after kingdom, phylum, class, order, family and genus. Humans, for example, belong to the genus *Homo* and the species *Homo sapiens*.

standardization

A process manufacturers may use to ensure batch-to-batch consistency of their products and to provide a measure of quality control. Dietary supplements are not required to be standardized in the United States. Some manufacturers use the term incorrectly or to mean different things and the presence of the word “standardized” on a supplement label does not necessarily indicate a level of product quality.

stimulant

A substance that increases brain activity, alertness, attention, blood pressure, heart rate, breathing, and energy.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

tea

A drink made by adding boiling water to fresh or dried herbs and steeping (soaking) them. It may be drunk either hot or cold. Also called an infusion.

therapeutic

Used to treat disease and help healing take place.

tincture

A liquid made by soaking an herb in a solution of alcohol and water. It is used for concentrating and preserving an herb and may be made in different strengths that are expressed as ratios of the weight of the dried herb to the volume or weight of the finished product.

tissue

A group or layer of cells in a living organism that work together to perform a specific function.

toxic

Having to do with poison or something harmful to the body. Toxic substances usually cause unwanted health effects.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

valerian

The roots of this plant are used by some cultures as an ingredient in mild sedatives and sleep aids for nervous tension and insomnia. It is being studied in improving sleep in patients undergoing treatment for cancer. Latin name: *Valeriana officinalis*.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

volume

The amount of space taken up by a substance; the amount of space a container can hold.

Updated: December 11, 2020 History of changes to this fact sheet